01_antonelli_vs_russell

May 12, 2026

1 Antonelli vs Russell — 2026 Qualifying, First Four Rounds

I wanted to see where Kimi Antonelli's 2026 season at Mercedes actually stands vs his teammate George Russell. This notebook compares both Mercedes drivers' fastest qualifying laps across the first four rounds of 2026, broken down by track segment. See the project README for full method and limitations.

1.1 Method (briefly)

For each race: load qualifying, take each driver's fastest valid lap, and segment the track from FastF1's `circuit_info.corners` (corners within 250 m are grouped, straights are filled in between). Telemetry — including FastF1's per-sample `Time` channel — is resampled onto a uniform 5 m distance grid. Per-segment time is read directly from that resampled `Time` channel as `Time[last_step] - Time[first_step]` inside the segment. Per-segment delta is `time(Russell) - time(Antonelli)`, so **positive = Antonelli faster**.

> An earlier version integrated `step / speed` per grid point. On 2 of 4 races the accumulated residual exceeded the 0.1 s sanity-check threshold (see the next section), so the analysis switched to reading FastF1's sample times directly. Residuals are now 0.1 s on all four races.

Each segment is classified into one of four categories by its minimum speed: slow corner (<130 kph), medium corner (130–200 kph), fast corner (>200 kph), or straight.

```
core           INFO     Loading data for Australian Grand Prix - Qualifying
[v3.8.1]

req            INFO     Using cached data for session_info

req            INFO     Using cached data for driver_info

req            INFO     Using cached data for session_status_data

req            INFO     Using cached data for track_status_data

req            INFO     Using cached data for _extended_timing_data

req            INFO     Using cached data for timing_app_data

core           INFO     Processing timing data…

core        WARNING     No lap data for driver 18

core        WARNING     No lap data for driver 55
```

```
core        WARNING     Failed to perform lap accuracy check - all laps marked
as inaccurate (driver 18)

core        WARNING     Failed to perform lap accuracy check - all laps marked
as inaccurate (driver 55)

req            INFO     Using cached data for car_data

req            INFO     Using cached data for position_data

req            INFO     Using cached data for weather_data

req            INFO     Using cached data for race_control_messages

core           INFO     Finished loading data for 22 drivers: ['63', '12', '6',
'16', '81', '1', '44', '30', '41', '5', '27', '87', '31', '10', '23', '43',
'14', '11', '77', '18', '3', '55']

core           INFO     Loading data for Chinese Grand Prix - Qualifying
[v3.8.1]

req            INFO     Using cached data for session_info

req            INFO     Using cached data for driver_info

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core           INFO     Processing timing data…

req            INFO     Using cached data for car_data

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core           INFO     Finished loading data for 22 drivers: ['12', '63', '44',
'16', '81', '1', '10', '3', '6', '87', '27', '43', '31', '30', '41', '5', '55',
'23', '14', '77', '18', '11']

core           INFO     Loading data for Japanese Grand Prix - Qualifying
[v3.8.1]

req            INFO     Using cached data for session_info

req            INFO     Using cached data for driver_info

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```
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'14', '18', '77', '11', '5', '6']
```

```
   Round                   Race  ANT lap (s)  RUS lap (s)  Δ lap (s)  ANT Q  \
0      1  Australian Grand Prix       78.811       78.518     -0.293      3
1      2     Chinese Grand Prix       92.064       92.286      0.222      3
2      3    Japanese Grand Prix       88.778       89.076      0.298      3
3      4       Miami Grand Prix       87.798       88.197      0.399      3

   RUS Q  Q mismatch
0      3       False
1      3       False
2      3       False
3      3       False
```

No Q-session mismatches across the 4 races.

1.2 Sensor quality

FastF1 telemetry doesn't carry an explicit quality flag, but a frozen `Speed` sensor produces a clear signature: only a handful of unique speed values across many samples within one segment. Any segment that fails this check is excluded from the headline category analysis below — the per-segment time delta is also unreliable in those segments because cumulative `Distance` is integrated from the bad speed.

```
Flagged 2 segment(s) where a driver's speed sensor looks frozen:

                 race segment      kind       category  min_speed_kph  \
0  Japanese Grand Prix      S8  straight       straight          189.0
1  Japanese Grand Prix      C8    corner  medium_corner          189.0

   delta_s_pre_filter
0              -1.663
1              -2.416
```

1.3 Sanity check

The segmentation logic is only useful if the segment deltas sum back to the actual lap-time delta. The table below verifies this on real telemetry: any |Δ| above 0.1 s would indicate a bug.

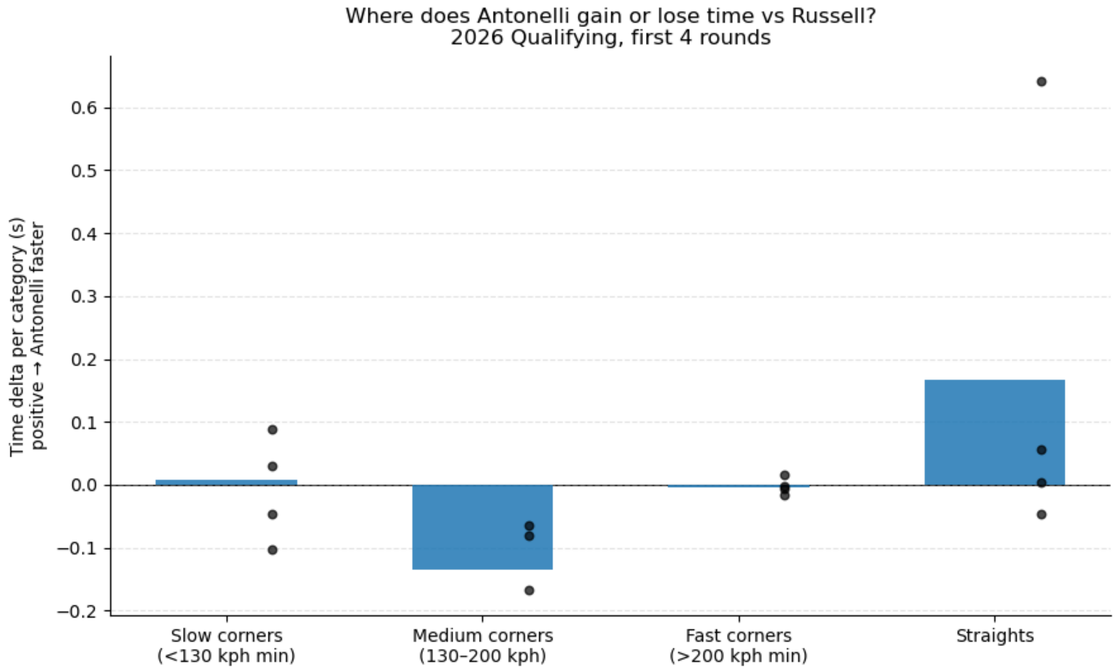
	Race	Σ segment Δ (s)	Lap-time Δ (s)	|residual| (s)
0	Australian Grand Prix	-0.3838	-0.293	0.09084
1	Chinese Grand Prix	0.2352	0.222	0.01318
2	Japanese Grand Prix	0.2758	0.298	0.02216
3	Miami Grand Prix	0.3758	0.399	0.02316

category event_name	fast_corner	medium_corner	slow_corner	straight
Australian Grand Prix	3	1	3	8
Chinese Grand Prix	2	1	5	9
Japanese Grand Prix	2	5	1	9
Miami Grand Prix	2	0	4	7

1.4 Headline finding — where does the time go?

Each bar is the mean per-category delta across all four races. Black dots show each individual race's per-category mean — they tell you whether the average reflects a consistent pattern or a noisy one.

```
Headline chart uses 60 segments after excluding 2 with sensor-freeze flags.
```



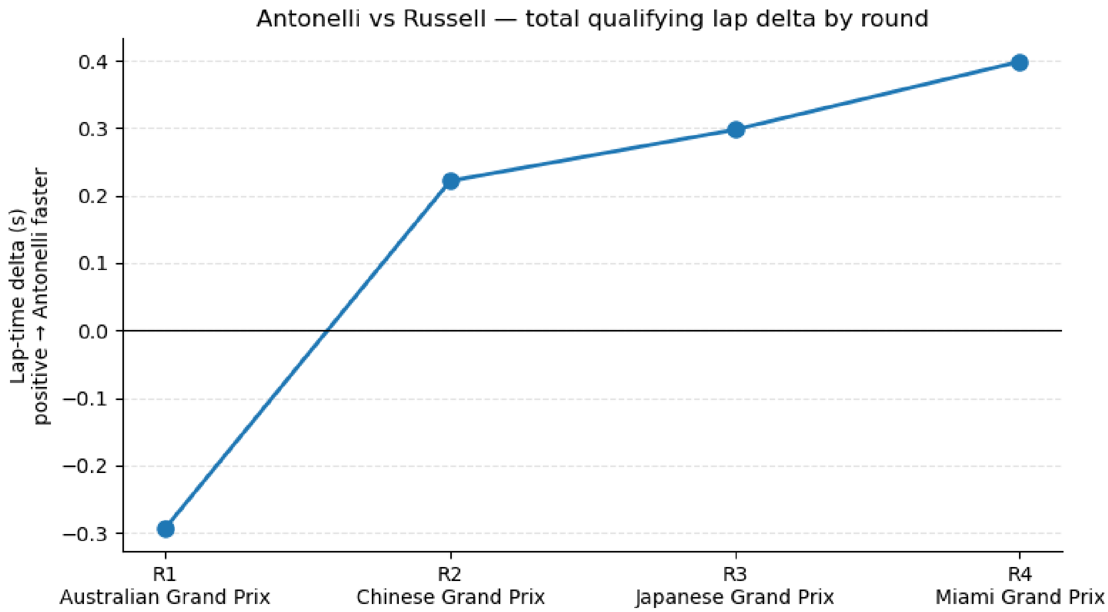

```
Per-category means after filtering (s/lap, positive = ANT faster):
category
fast_corner     -0.0034
medium_corner   -0.1355
slow_corner      0.0088
straight         0.1660
```

Antonelli is **faster than Russell on net** across rounds 2–4, and the category breakdown after the sensor-quality filter tells a cleaner story than the raw average suggested:

- **Straights:** Antonelli gains **+0.17 s/lap** — the largest and most consistent positive across all four rounds.
- **Fast corners (>200 kph):** essentially tied (**−0.003 s/lap**).
- **Slow corners (<130 kph):** essentially tied (**+0.009 s/lap**, effectively noise).
- **Medium corners (130–200 kph):** Antonelli is slightly slower at **−0.14 s/lap** — a real but small deficit, much smaller than the −0.46 s/lap that an unfiltered average produced.

The unfiltered medium-corner number was inflated almost entirely by a single Japan segment where Antonelli's speed sensor froze at 189.0 kph for a long stretch (see the "Sensor quality" diagnostic above the chart). Once the two affected segments are excluded, Antonelli's straight-line advantage emerges as the clearer headline and the medium-corner gap shrinks to a more credible size.

1.5 Trajectory — is the gap closing?



Across four rounds, the lap-time gap has gone from **−0.29 s** (Russell faster by 0.29 s) at Australia to **+0.40 s** (Antonelli faster by 0.40 s) at Miami — a monotone trend in Antonelli's favour covering **0.69 s across four races**. Four data points is too few to project a trajectory with confidence, but the direction is striking: Antonelli was behind at Round 1 and ahead on every subsequent round, with the margin growing each time. Australia looks like the outlier round, not the norm.

1.6 Where on the track is each driver gaining time?

The category breakdown averages across each segment, but it can hide where exactly within a segment the time is found. Below, each circuit is shown as a track map coloured by the **local slope of the cumulative time delta** — the rate at which the gap is changing as the lap unfolds.

- **Blue** = Antonelli is gaining on Russell at that part of the track.
- **Red** = Russell is gaining on Antonelli.
- Corner numbers (from FastF1's `circuit_info`) are overlaid for orientation.

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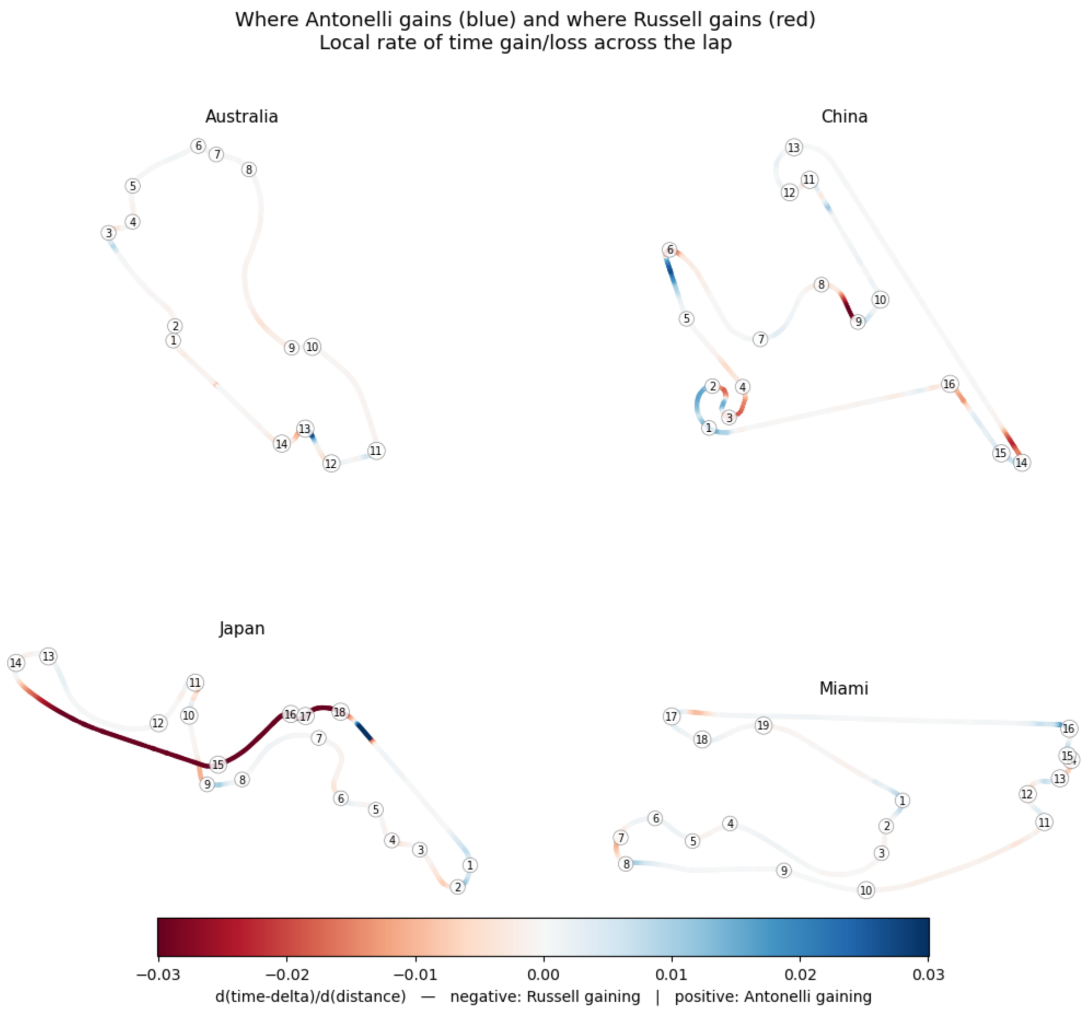
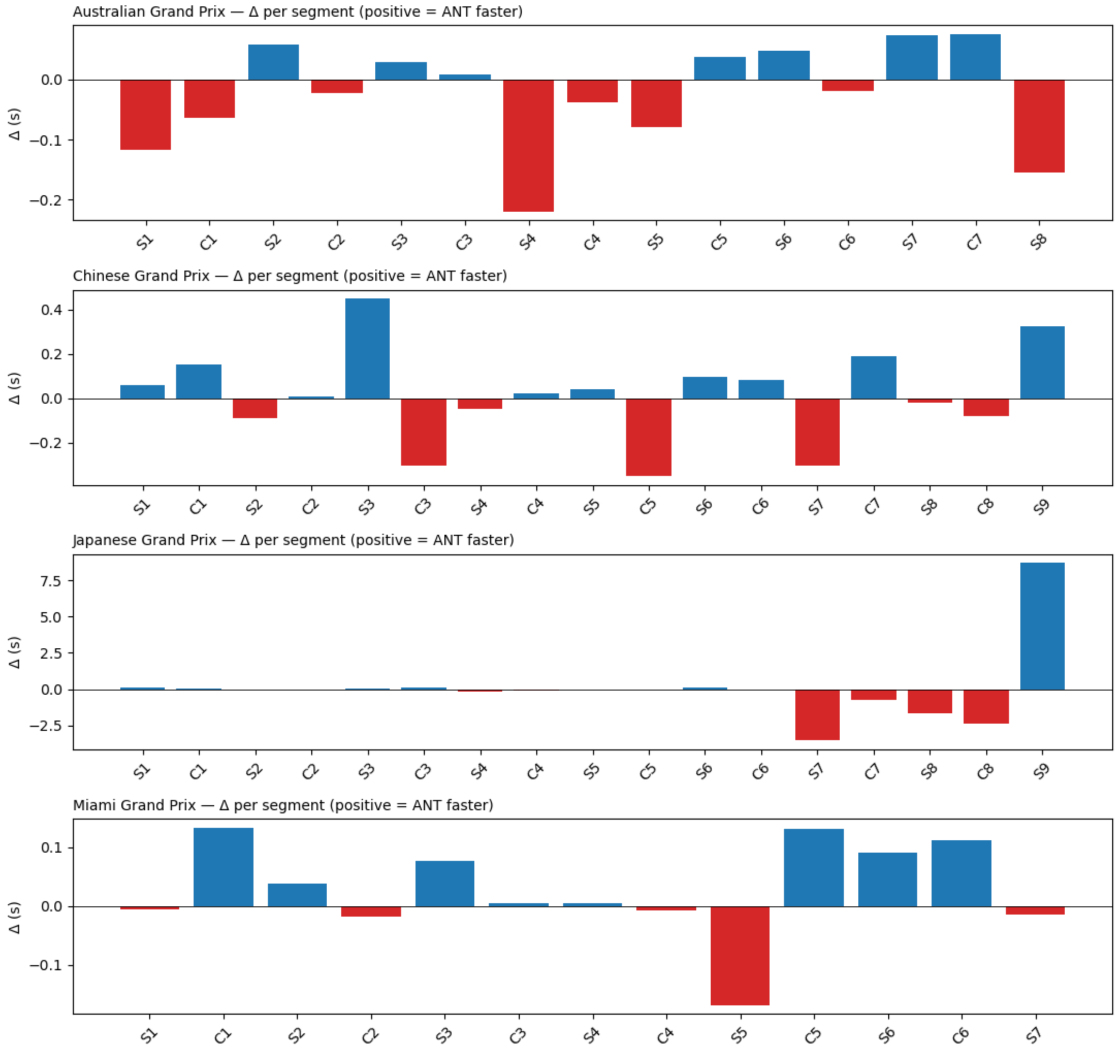


1.7 Per-race detail (appendix)

For each race, the raw per-segment delta. Useful for curious readers; not the main story.



1.8 Caveats

- **Sample size: four races.** Findings are directional, not conclusive.
- **Q-session timing.** Q1 / Q2 / Q3 happen on an evolving track. The summary table flags any race where the two drivers' fastest valid laps came from different Q-segments; treat the delta on those races with extra skepticism.
- **Setup divergence.** Mercedes drivers do not always run identical setups. Public telemetry can't separate driver from setup.
- **Traffic and tire age within Q.** Out-laps, in-laps, and intra-session tire age all affect achievable lap time.
- **Telemetry sensor freezes.** As shown above, FastF1's car telemetry can stop reporting changes for long stretches (Antonelli's Japan lap is the clearest case in this dataset). When that happens, integrated `Distance`, `Speed`, and the X/Y trajectory all become unreliable in the affected segments. Lap-level and sector-level time deltas are unaffected — those come from timing-line beams, not from the car. Affected segments are filtered out of the category

headline above; expect mild visual distortion at Japan's T14–T18 in the track-delta map.

These caveats are why the README frames this as a "careful look at what the telemetry shows," not a verdict on relative driver skill.

1.9 What I found

After working through the four-race dataset, the cleanest read is this: Antonelli is **faster on straights** (+0.17 s/lap on average, the strongest signal in the data), and essentially level with Russell in fast corners and slow corners. He's **modestly slower in medium corners** (−0.14 s/lap), but the deficit is much smaller than the raw average first suggested — once I filtered out two Japan segments where Antonelli's speed sensor was frozen for ~1300 m of the lap, the medium-corner gap shrank from −0.46 to −0.14 s/lap.

The lap-level findings stay the same: Antonelli was 0.29 s slower at Australia (R1) but has been faster than Russell every round since, with the margin growing each race (R2 +0.22 → R4 +0.40). Sector-time data — which comes from timing-line beams, independent of the car telemetry — confirms each of those lap-level deltas, including Japan, so the trajectory finding is robust to the sensor issues.

My read: Australia was the outlier, not the norm. Antonelli's strength so far is **single-lap pace on straights and high-speed sections** — exactly where outright commitment and brake-late confidence translate to time — while medium-speed corners are the one area where Russell still has a small edge. I'd want to see more rounds, and ideally clean telemetry at every circuit, before drawing stronger conclusions.

I'll re-run the same framework after each race for the rest of 2026 — see What I'd build next for the extensions.